Atul

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PROFESSIONAL SUMMARY

Technology undergraduate with a strong foundation in machine learning, generative AI, and full-stack application development. Brings practical experience in building data-driven solutions, working with cloud platforms, and translating real-world requirements into functional products. Combines an analytical, research-oriented mindset with strong problem-solving ability and a focus on creating reliable, user-centric technology.

EXPERIENCE

Astrea IT Services Pvt. Ltd. **June 2026–July 2026**
Salesforce Development Intern *Remote*

- Configured CRM components and developed cloud-based solutions on the Salesforce.com platform for enterprise use cases.
- Completed hands-on development exercises covering Salesforce workflows, cloud application development, and enterprise platform practices.
- Applied CRM configuration concepts through practical assignments, improving understanding of data management, process automation, and user-focused Salesforce solutions.

Zidio **Remote**
Machine Learning Engineer Intern *3 Months*

- Contributed to Project FORESIGHT, an end-to-end AI/ML and full-stack platform for product-demand forecasting and inventory intelligence.
- Developed forecasting and analytics features to identify stockout and overstock risks and deliver AI-powered inventory recommendations through an interactive dashboard, supporting data-driven supply-chain decisions.
- Integrated machine learning outputs with dashboard workflows to present forecasts and inventory insights in a clear, actionable format for business users.

PROJECTS

Crop Recommendation System | *Python, Flask, React, Scikit-learn, Pandas, NumPy, Machine Learning*

- Developed a Flask-based machine learning web application that analyzes soil nutrients and environmental conditions—including nitrogen, phosphorus, potassium, pH, temperature, humidity, and rainfall—to recommend suitable crops for cultivation.
- Built and integrated a Scikit-learn prediction pipeline using Pandas and NumPy for data preprocessing and model inference, and connected it to a responsive React interface that presents crop and fertilizer recommendations to users.

MyCity Help | *HTML, CSS, JavaScript, Python (Flask), MongoDB, OpenAI API*

- Developed a full-stack Progressive Web Application for civic issue reporting and management, enabling citizens in Kanpur to submit municipal complaints, monitor their status, and track issues through resolution.
- Integrated AI-powered chatbot assistance and image-based issue categorization alongside interactive Leaflet maps, Chart.js analytics, and gamification features such as points, badges, and leaderboards to improve citizen engagement.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript
Backend: FastAPI, Flask, REST APIs.
Frontend: HTML, CSS, Bootstrap, JavaScript.
Databases: MongoDB, MySQL, Supabase.
Tools & Platforms: VS Code, Google Colab, OpenAI APIs, Git, GitHub, Latex
Machine Learning: NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, OpenCV, RAG, Deep Learning, LLM, Agentic AI.
Core CS: Data Structures & Algorithms, Database Management Systems, Computer Networks, Operating Systems

RESEARCH EXPERIENCE

Federated Generative Intelligence for Explainable and Autonomous Cyber Defence in Critical Infrastructures
IEEE ISCS 2025 | DOI: 10.1109/ISCS69371.2025.11386415

- Designed and evaluated a privacy-aware cyber-defence architecture combining Federated Learning, Generative AI, SHAP explainability, and Deep Reinforcement Learning to enable autonomous threat detection on the SWaT dataset without centralizing sensitive operational data.

Agentic AI Framework for Autonomous and Self-Managing Cloud Services
IEEE SCEECS 2026 | DOI: 10.1109/SCEECS68810.2026.11429932

- Developed a Multi-Agent Reinforcement Learning framework integrated with Kubernetes orchestration to coordinate adaptive resource allocation, automated recovery, and workload management, improving service responsiveness and operational resilience.

EDUCATION

Allenhouse Institute of Technology **Kanpur**
Bachelor of Technology *2023–2027*

B.G.V. Inter College **2022**
Intermediate (Class XII)

ACHIEVEMENTS

- Secured Rank 7,007 in the TCS CodeVita global coding competition.
- Achieved AIR 7,361 and ranked in the top 20% in NCAT 2026.
- Second Runner-Up at Idea Sprint 2025 for the MyCity Help smart-city solution.

TRAININGS & CERTIFICATES

- **Full Stack Development Training** — Allenhouse Institute of Technology (2026).
- **Agentic AI Certification and Training** — Oracle, Infosys (2026).
- **Data Science Training** — Apna College (2026).
- **Artificial Intelligence and Generative AI Training** — IIT Kanpur (2025).
- **Python Certification** — Coding Ninjas (2025).